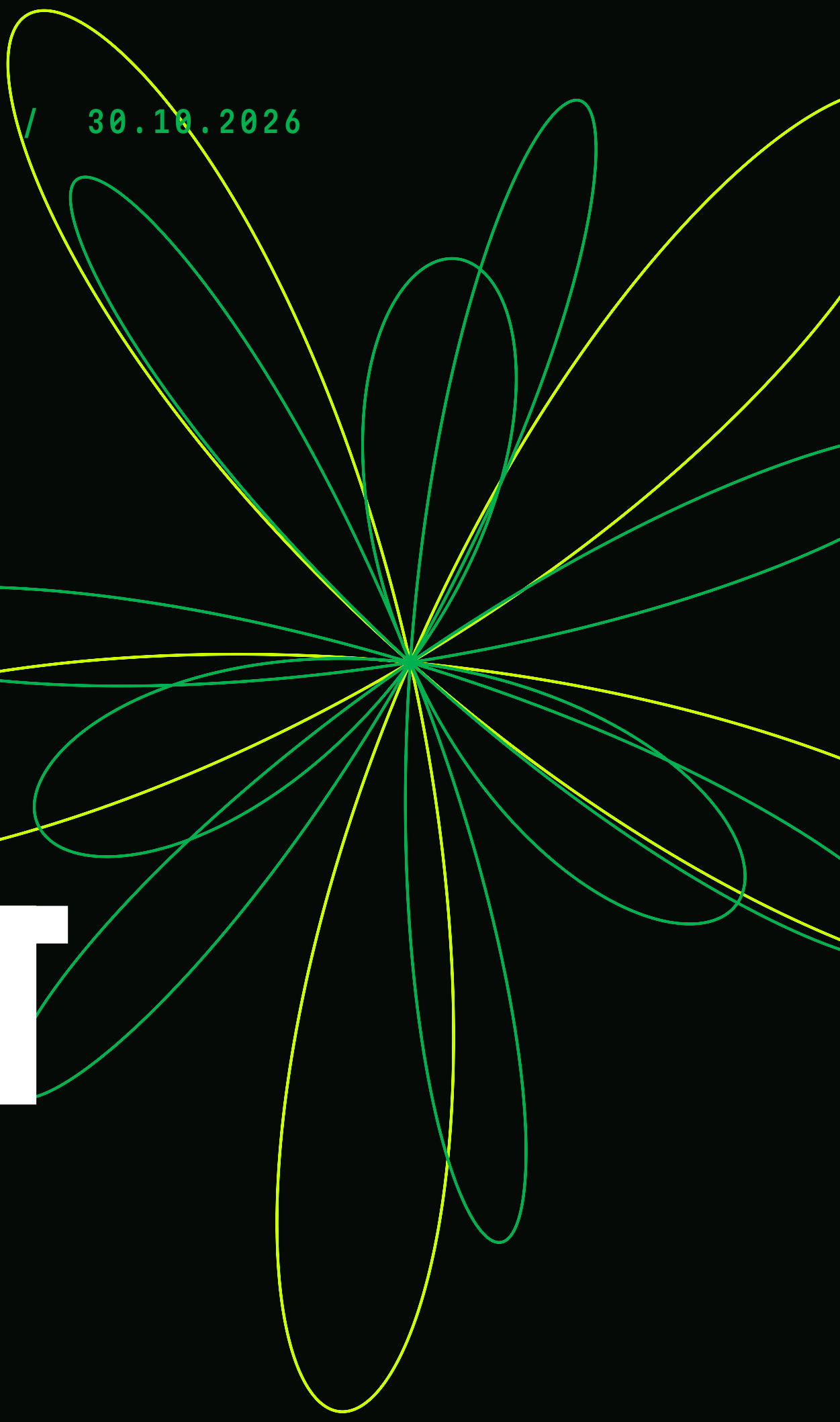


THE NIGHT PIN

Six layers of data anyone can buy. One join gives them a name.

18.03.2026 / US SENATE / UNDER OATH

"We do purchase commercially available information ... and it has led to some **valuable intelligence** for us."

KASH PATEL, FBI DIRECTOR, TO SEN. RON WYDEN[1]

Asked to commit to not buying Americans' location data, he did not. The FBI holds that it needs no warrant for data it pays for. No court has tested that theory.[1]

THE STACK

SIX LAYERS

06	**IDENTITY**	**Voter files** the name on the door
05	**DEVICE**	**Ad IDs from ad auctions** where a phone sleeps and goes
04	**ORBIT**	**Satellite imagery** cars, tanks, ships
03	**ATTENTION**	**App and web panels** what people open
02	**NETWORK**	**NetFlow** who talks to whom
01	**STRUCTURE**	**Shodan** machines on the internet

1 TO 4 WATCH THINGS. 5 AND 6 WATCH PEOPLE.

01 STRUCTURE

SHODAN

A search engine that crawls the internet and indexes every device that answers: servers, webcams, routers, industrial controllers. It maps machines, not people. [6]

---

02 NETWORK

NETFLOW

90%

Traffic metadata: which address talked to which, when, how much. Team Cymru's Augury draws on 550 collection points and adds 100 billion records a day. Contract records put its reach near 90 percent of global traffic. Buyers included several branches of the US military. [3]

ONE FLOW RECORD / ILLUSTRATIVE, DOCUMENTATION ADDRESSES

```
SRC 203.0.113.7  DST 198.51.100.24
PORT 443  BYTES 18,442  T 14:02:11
NO CONTENT. JUST WHO, WHERE, WHEN, HOW MUCH.
```

03 ATTENTION

THE PANEL

Panels estimate what people download, open and stream. Funds buy the estimates to see a quarter before the company reports it.

14.09.2021 / SEC / FIRST ALTERNATIVE-DATA FRAUD CASE

$10,000,000

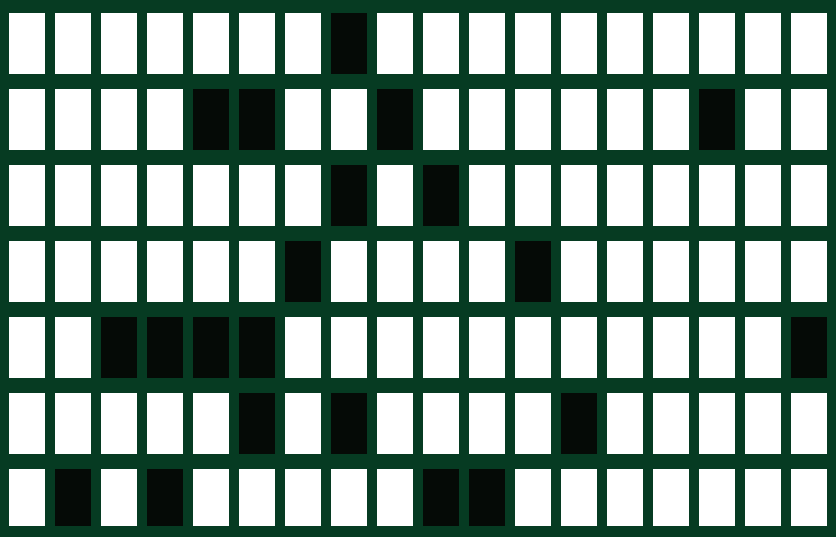
App Annie told companies their app data would be aggregated and anonymized. From 2014 to 2018 it used the raw data to tune estimates it sold to trading firms. Its founder paid $300,000 and took a three-year officer bar. [4]

04 ORBIT

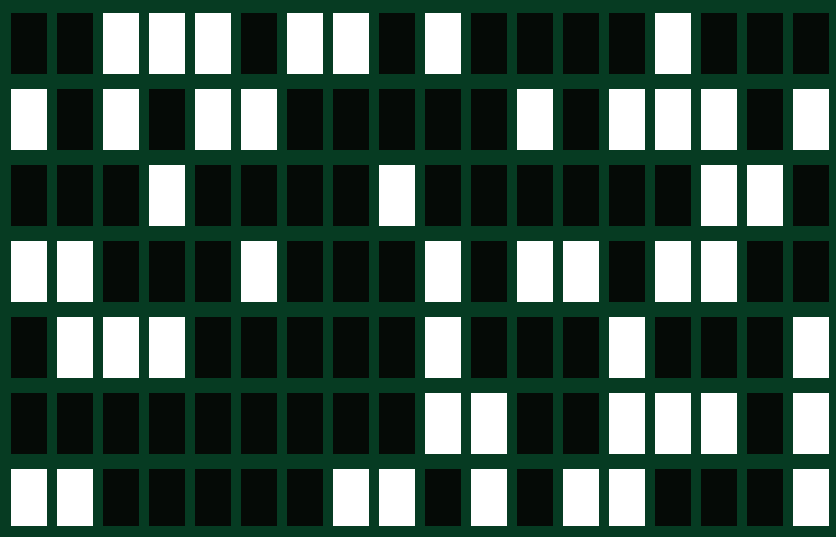
4 to 5%

Earned in the three days around earnings by funds that counted cars in retailers' parking lots from orbit. Berkeley Haas: 4.8 million images, 67,000 stores, 44 retailers, 2011 to 2017.[5]

QUARTER A / 82% FULL



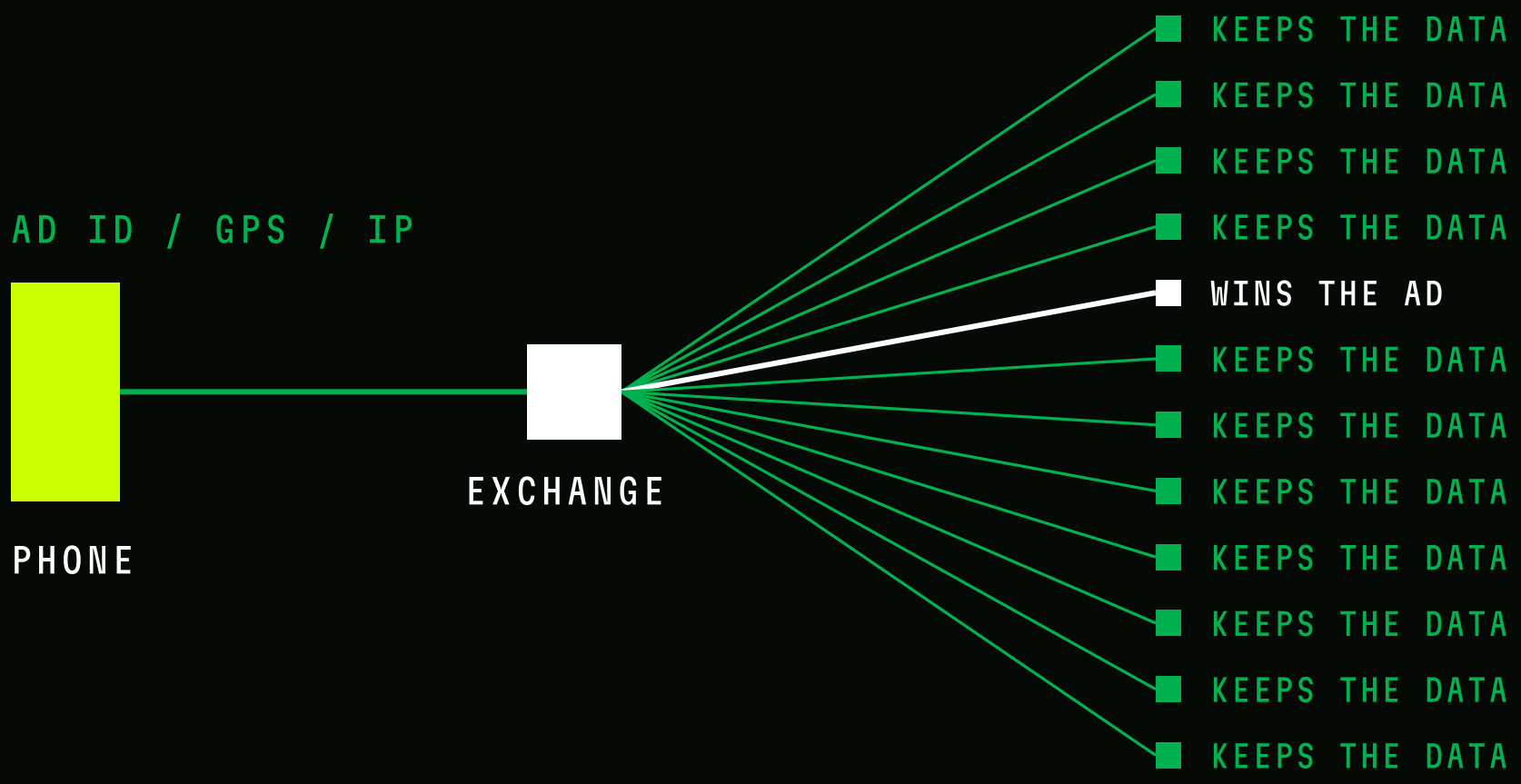
QUARTER B / 41% FULL



The edge was biggest on the short side. Individual investors were buying the same stocks the funds were betting against.[5]

06 ILLUSTRATION: TWO LOTS, TWO QUARTERS

EVERY AD IS AN AUCTION



When an app shows an ad it broadcasts a bid request: advertising ID, GPS when permitted, IP address, device, app. Every bidder receives it, including the ones that lose. [2]

1,000,000,000

Phones Gravy Analytics and Venntel say they draw on, at 17 billion signals a day.[7]

500,000,000+

Ad IDs Mobilewalla matched to precise location, 2018 to 2020. An estimated 60 percent came from ad auctions.[7][2]

30,000,000+

Location points in one leaked Gravy sample, January 2025. Devices at the White House, the Kremlin, the Vatican and military bases.[8]

NOT IN THE BID REQUEST

CARD PURCHASES

Transaction data is a separate product from card and bank panels. The auction does not carry it.[2]

A GUARANTEED IPHONE ID

Tracking has been opt-in since 2021. Measured opt-in runs about 35 to 50 percent.[9]

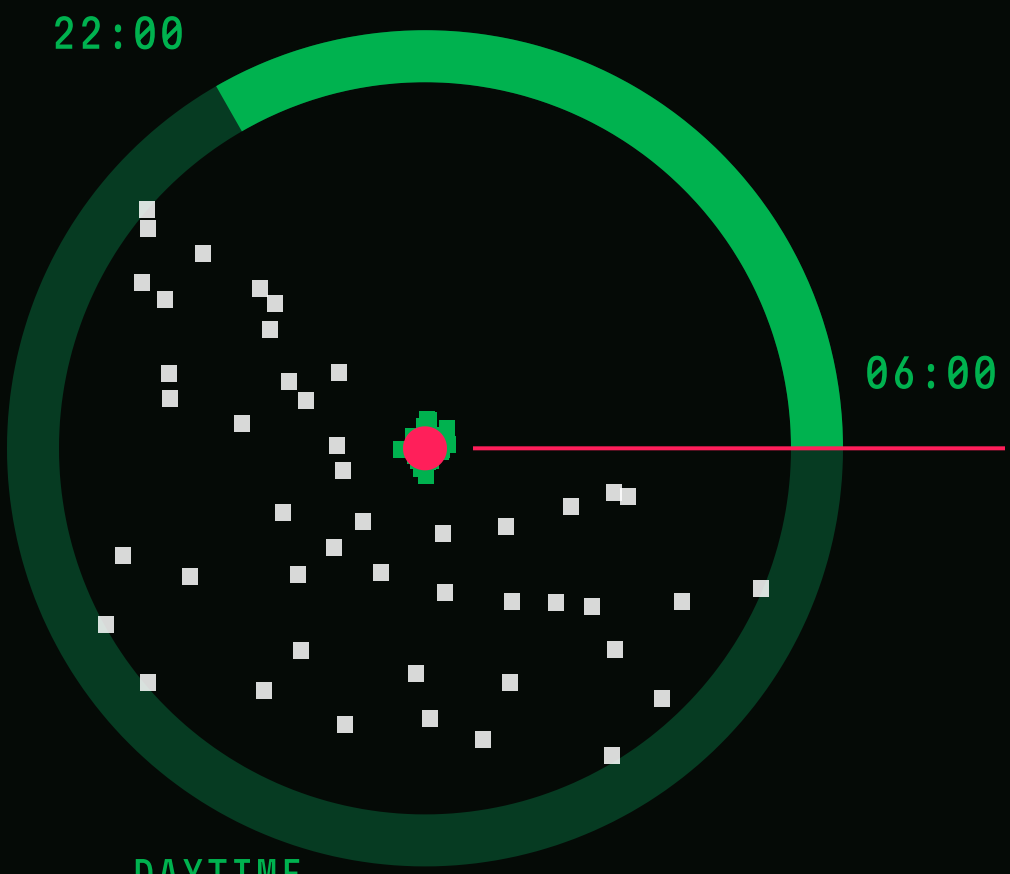
ALWAYS GPS

Much of the leaked Gravy data was inferred from IP addresses.[8]

BUT ANDROID

Google ended Privacy Sandbox in October 2025. The ad ID stays on unless deleted.[10]

22:00 TO 06:00



ONE LOOKUP

HOME COORDINATE

STREET ADDRESS

NAME

YEAR OF BIRTH

PARTY

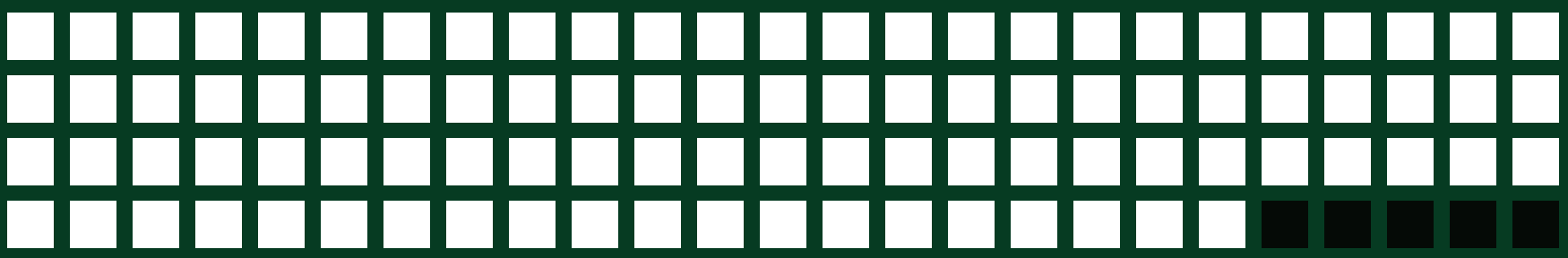
TURNOUT HISTORY

A phone that rests at one coordinate every night has a home. A home has an address. An address has registered voters. That single lookup is where an ad ID becomes a person.

PROOF / SCIENTIFIC REPORTS, 2013

4 → 95%

Four points in place and time singled out 95 percent of 1.5 million people in a mobile dataset, even at hourly, cell-tower precision. Anonymous traces are not anonymous. [11]

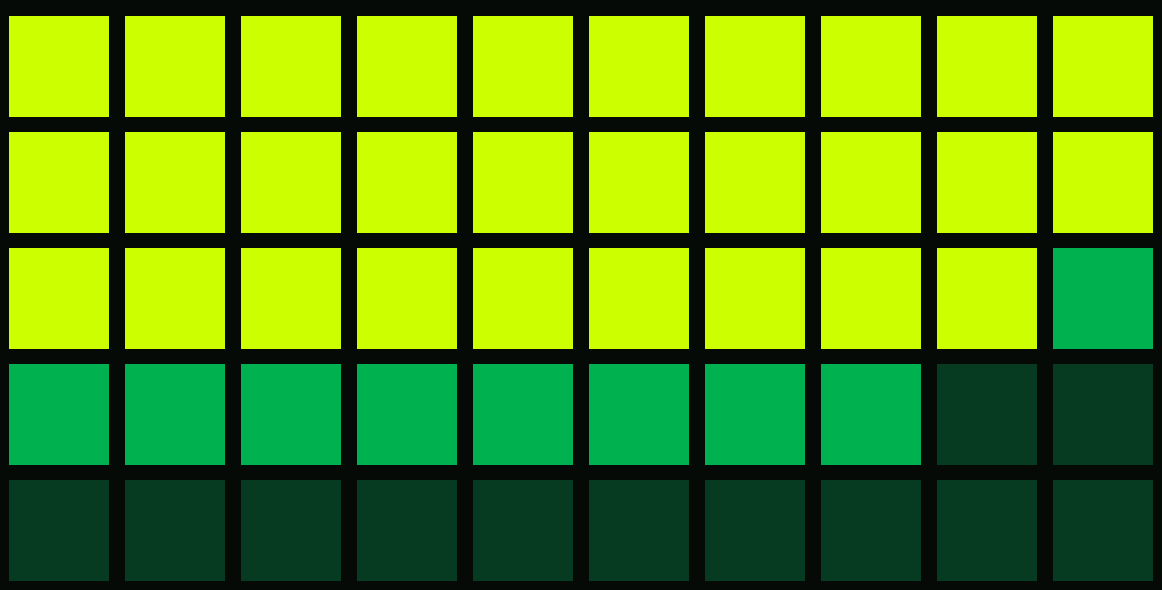


NEW YORK TIMES / 2019

Reporters took 50 billion pings from 12 million phones and followed officials, lawyers and police by name. One Microsoft employee visited an Amazon office and took a job there a month later. [15]

NAME. ADDRESS. PARTY. TURNOUT.

Almost every state file carries name and home address. Many add party, voting history and year of birth.[12]



29
NON-COMMERCIAL ONLY

9
ELECTION OR GOVERNMENT

12
SILENT ON USE

No state's rules expressly allow any use. A trading desk is a commercial user. The pitch skips this.[12]

POLITICS

On 18 June 2026, L2 and Nexxen launched VoterMatch, joining L2's national voter file to ad-auction logs so campaigns can see which voters an ad reached and how often. The matching method and match rates are not disclosed.[13]

VOTER FILE

NAME
ADDRESS
PARTY
TURNOUT

AD-AUCTION LOG

AD ID
IMPRESSION
TIME
APP

JOIN

REACH + FREQUENCY, PER VOTER

17.06.2020 / LAYERS 5 AND 6, DEMONSTRATED

Mobilewalla published the race, gender and age mix of protesters in New York, Los Angeles, Minneapolis and Atlanta, from phone data. Its founder said no agency asked for it. The FTC later barred it from selling sensitive location data.[19][7]

ALPHA?

01 FOOT TRAFFIC / WORKS

Sold as store-level counts, not people. In 2019 Thasos offered quant funds daily visits to 600,000 US properties. [14][5]

02 NAMED PEOPLE / BARRED

FTC orders bar the big brokers from selling sensitive location data, including where people live. Trading on data obtained by breaking a promise is the App Annie case. [7][18][4]

03 VOTER BLOCKS / UNPROVEN

38 states limit the input. No published evidence it beats polling.[12]

NOT FUNDS. STATES.

$27,000,000

FBI contract ceiling for 5,000 Locate X licenses, 2022. Locate X runs on Venntel data.[7]

DHS 113,654 location points in three days, 2018.[7]

CBP Ad-auction location pilot, 2019 to 2021.[2]

ICE Bought Webloc in 2025. It filters millions of phones by ad ID.[2]

"CAI includes information on **nearly everyone**."

CAI is commercially available information. The same report says it can be used to harm a person's reputation, emotional well-being or physical safety. [16]

THE LAW, AS OF TODAY

29.06.26

Chatrie v. United States. Getting a phone's location history from Google is a Fourth Amendment search.[17]

NOT DECIDED

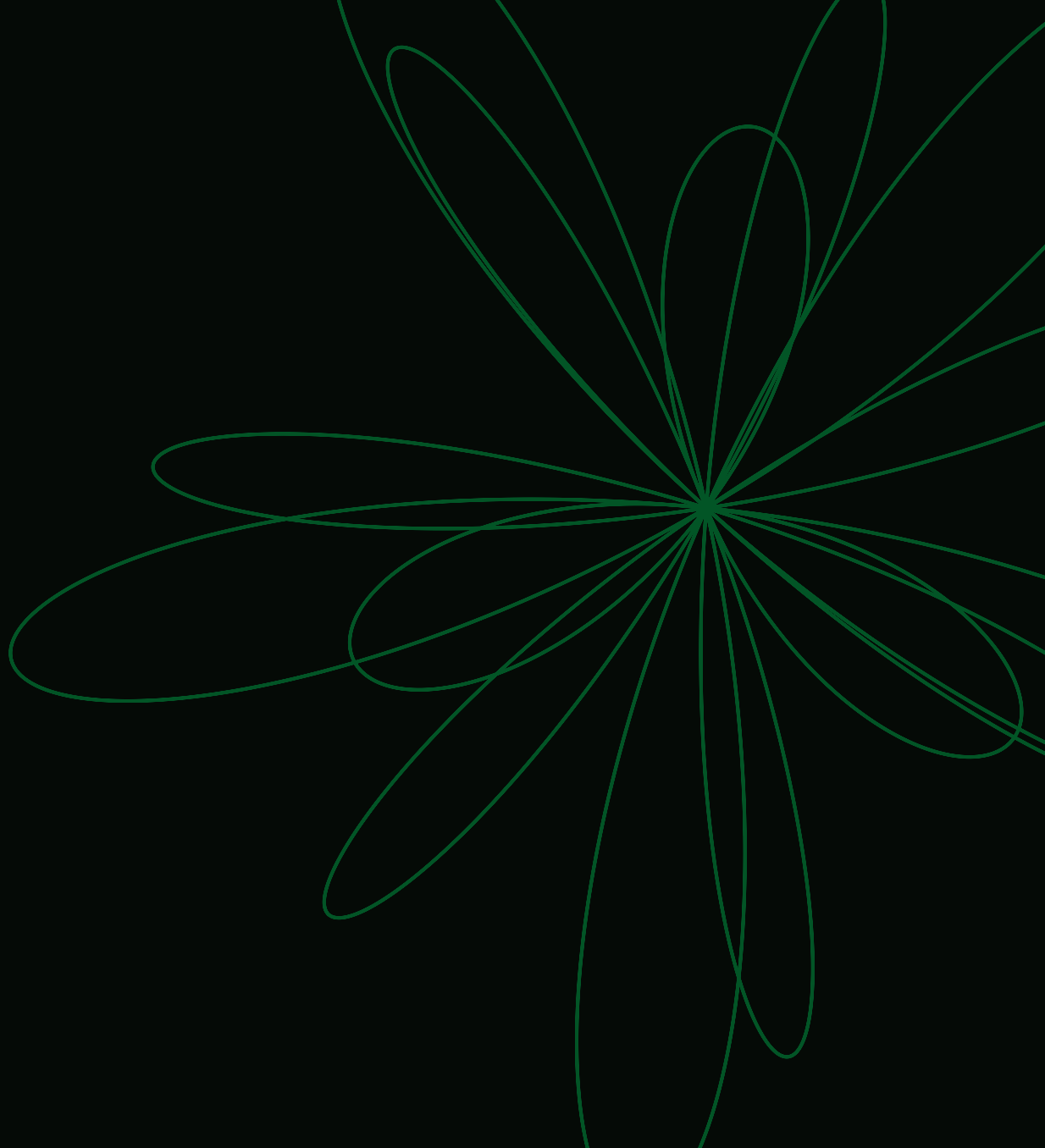
Data the government buys. The FBI's no-warrant theory is untested.[1]

PENDING

The Government Surveillance Reform Act would require a warrant before federal agencies buy Americans' data from brokers.[1]

SETTLED

Kochava order, filed May 2026: no sale of sensitive location data without express consent.[18]



Chatrie put a warrant in front of the data police ask for.

Nothing stands in front of the data they buy.

THE CHEAPEST WARRANT IS A PURCHASE ORDER.

DELETE THE ID

ANDROID

Settings, Privacy, Ads, Delete advertising ID. The path varies by maker.

IPHONE

Settings, Privacy & Security, Tracking. Allow Apps to Request to Track: off.

EVERY APP

Location only while using, approximate where possible.[2]

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